

ADI TAUQIR

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EDUCATION

Georgia State University

Atlanta, GA

B.S. and M.S. in Computer Science | GPA: 4.25

May 2028

- President's List: Fall 2024 – Spring 2026 | **Coursework:** Data Structures & Algorithms, Computer Organization, Mobile Dev, Artificial Intelligence, Data Mining, Cloud Computing

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++ , Swift, SQL, HTML/CSS, Kotlin, Flutter, Bash/Shell

Software Engineering: React, Next.js, Node.js, FastAPI, SwiftUI, REST APIs, WebSockets, Object-Oriented Programming, Data Structures & Algorithms, Unit Testing, Regression Testing, Debugging, Performance Optimization

Data & AI: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Machine Learning, Predictive Analytics, Feature Engineering, Model Evaluation, Generative AI, RAG, AI Agents, Workflow Automation, Structured Outputs

Infrastructure & Tools: PostgreSQL, Supabase, SQLite, Redis, Docker, Linux, AWS EC2, Azure, Git, GitHub Actions, CI/CD, Tailscale, Cursor, Codex, Claude Code

TECHNICAL PROJECTS

Kaiko – Adaptive Agent Control Plane | TypeScript, Node.js, JSON-RPC, LLM Agents

Aug. 2026 – Present

- Built an **agentic AI control plane** that coordinates local models, tool execution, search, and delegated subagents through bidirectional JSON-RPC IPC, streaming events, persistent state, and human approval workflows rather than treating an LLM as a single black-box request.
- Designed a quantitative decision layer using **Rasch 1PL Item Response Theory, Bayesian MAP estimation, Newton–Raphson optimization, and Fisher Information** to continuously estimate user state and choose the next interaction dynamically.
- Instrumented the runtime with telemetry for latency, accuracy, confidence, standard error, and interaction history, creating reusable signals for **model evaluation, analytics, experimentation, and performance monitoring**.
- Root-caused a recurring **2–4 second latency stall** to repeated interactive-shell binary discovery and replaced it with memoized lookup, turning repeated path resolution into an effectively immediate in-memory operation.

AWRY – Recession Nowcast & Risk Engine | Python, Scikit-learn, Pandas, FRED/ALFRED, Streamlit

Jan. 2026 – Apr. 2026

- Turned **13 heterogeneous macroeconomic indicators** into a forward-looking recession signal by building an end-to-end data pipeline for ingestion, temporal alignment, cleaning, feature engineering, training, evaluation, and inference.
- Engineered lagged, differenced, spread, and forward-shifted features to expose leading structure hidden in raw time-series data; trained Logistic Regression and Random Forest models reaching **0.994 AUROC, 0.86 F1, and 0.024 Brier score**.
- Backtested model behavior across the structurally different **2001, 2008, and 2020 recession regimes**, using historical failure modes to analyze calibration, false positives, signal persistence, and generalization under changing conditions.
- Converted probabilistic model outputs into a Streamlit **decision-support system** that surfaces forecast confidence, indicator movement, and model signals for faster technical and business interpretation.

FYouTube Premium – Runtime Compatibility Layer | JavaScript, WebKit, Browser APIs, GitHub Actions

Aug. 2026 – Present

- Reverse-engineered YouTube's changing client-side behavior and built a cross-browser compatibility layer for Orion/WebKit, Chrome, and Firefox, shipping **16+ user-facing features across 56+ commits**.
- Treated the browser as an **unreliable distributed runtime**: handled asynchronous DOM mutations, playback-state races, navigation changes, browser API differences, and upstream UI breakage through event-driven JavaScript and defensive state management.
- Built regression checks, browser-specific packaging, version validation, and **GitHub Actions CI/CD** to make testing and releases repeatable across three independently behaving browser environments.
- Used external user reports as a lightweight production feedback loop, reproducing failures on real devices and iterating on **reliability, compatibility, performance, and customer-facing behavior**.

PROFESSIONAL EXPERIENCE

Regal Cinemas

Atlanta, GA

Floor Staff

Aug. 2025 – Present

- Resolve customer, transaction, and operational issues in a high-volume environment while processing **30+ transactions per shift with zero cash discrepancies** and coordinating with teammates under time constraints.

LEADERSHIP & ACHIEVEMENTS

HackHers – Best Use of Gemini Award (Mosaic) | Gemini API, Agentic AI, SwiftUI

Sept. 2026

- Built an agentic financial assistant orchestrating multi-step Gemini workflows and context harnesses for messy financial data.

CodePath TIPS 102 – Advanced Data Structures & Algorithms Certificate

Dec. 2025